

Jericho Down Equalizer Guide

Plain-English help for improving worship, speech, podcast, and live vocal audio without needing an audio engineering degree.

Quick Rule

Make the smallest change that solves the problem. If a control makes the sound better at first but worse after ten seconds of listening, back it down.

A Good Starting Workflow

Step	What To Do
1. Pick a preset	Start with Flat for truth, Podcast Clean for safe speech, Warm Radio or Deep Warm for bigger broadcast tone, Noisy Room for problem rooms, or Bright Headset for dark headset mics.
2. Set input trim	Talk at real volume. Peaks should be healthy but not clipped. If it clips before DSP, no effect can fully repair it.
3. Fix low-end problems first	Use high-pass and de-popper before boosting warmth. Cutting junk is cleaner than compressing junk.
4. Shape with EQ	Use small moves. 1-3 dB is often enough. Sweep only when hunting for a problem frequency.
5. Control dynamics	Use gate/expander for background noise, compressor for steady loudness, limiter for final safety.
6. Listen in context	A solo mic can sound impressive but fail in a mix. Re-check while music or other mics are active.

Built-In Presets

Presets are starting points. Pick the closest sound, then make small EQ and DSP changes for the room, mic, and singer or speaker.

Preset	Plain-English Use
Flat	Neutral reference. Start here when you want to hear the mic without a voice style.
Podcast Clean	Cuts rumble and mud, adds presence for clearer speech, and keeps volume steady.
Warm Radio	Adds body and warmth, gently tames harshness, and gives a denser broadcast feel.
Deep Warm	Leans into chest and low-mid body, carves mud, softens bite, and uses dense compression.
Noisy Room	Raises the high-pass filter and tightens the gate to fight room noise.
Bright Headset	Adds clarity for darker headset mics while trimming low rumble and high fizz.

The EQ Sliders

The equalizer changes tone by frequency. Left sliders affect low sounds; right sliders affect high sounds. When in doubt, cut problem areas before boosting favorite areas.

Zone	Approx. Range	What You Hear	How To Adjust
Rumble	31-90 Hz	Desk thumps, stand bumps, plosives, HVAC rumble.	Usually cut. Raise high-pass if the room shakes or P sounds hit hard.
Warmth	125-250 Hz	Chest, fullness, weight.	Add a little for thin voices. Cut a little if the voice sounds thick.
Mud	250-500 Hz	Cloudy, boomy, blanket-over-the-mic tone.	Cut gently to clear speech or vocals. This zone gets ugly fast if boosted.
Box/Nasal	500 Hz-1.4 kHz	Hollow box sound, honk, nasal tone.	Small cuts can make cheap rooms and headset mics sound more natural.
Presence	2-5 kHz	Words become easier to understand.	Boost gently for clarity. Too much gets harsh or tiring.
Sibilance	5-9 kHz	S, sh, hiss, sharp bite.	Cut or use de-esser if S sounds jump out.
Air	10-20 kHz	Breath, sparkle, hiss.	Add only if the mic sounds dull. Cut if hiss or fizz is obvious.

Fast EQ Moves That Usually Work

- Rumble or P pops: cut 31-90 Hz or raise the high-pass filter.
- Muddy voice: cut a little around 250-500 Hz.
- Boxy or nasal voice: cut a little around 500 Hz-1.4 kHz.
- Needs clearer words: add a little 2-5 kHz presence.
- S sounds hurt: cut 5-9 kHz or use de-esser.
- Too dull: add a little air above 10 kHz, but stop before hiss gets louder.

DSP Effects

DSP effects process level, noise, tone, and safety. The order matters in the app: clean the input, shape tone, control dynamics, then protect the final level.

Effect	What It Does	Main Controls
Input Trim	Sets the level before DSP. This is the first gain stage.	Preamp: raise if the mic is too quiet; lower if peaks clip.
De-popper	Reduces low-frequency blasts from P/B sounds.	Pop reduction: strength. Frequency: low band to watch. Threshold: how loud a pop must be before action.
High-pass Filter	Removes rumble below the chosen frequency.	HPF Hz: raise for more cleanup; lower if the voice gets thin.
Hum Removal	Notches electrical hum and its low harmonic.	Frequency: use 60 Hz in the US, 50 Hz where power is 50 Hz. Amount is implied by the filter design.
Noise Gate	Turns down the mic when the signal falls below a threshold.	Threshold: when gate closes. Attack: open speed. Hold: stay-open time. Release: close speed. Range: how far it turns down.
Expander	A gentler gate that reduces room noise without hard cutoffs.	Threshold: when reduction starts. Ratio: strength. Range: max reduction. Attack/Hold/Release: timing.
Noise Suppression	Reduces steady background noise such as fans.	Amount: more cleanup. Sensitivity: how aggressively it identifies noise. Too much can sound watery.
Echo Reducer	Tames room reflections and slap echo.	Amount: reduction strength. Sensitivity: how easily it reacts to echo-like tails.
Compressor	Smooths loud and quiet parts so the mic stays present.	Threshold: level where compression starts. Ratio: strength. Attack: catch speed. Release: recovery. Knee: softness.
De-esser	Controls sharp S and sh sounds.	Amount: reduction strength. Frequency: sibilance target. Threshold: when it reacts. Range: max reduction.
Breath Reducer	Turns down airy breath noise between phrases.	Amount: strength. Sensitivity: how easily it identifies breath.
Presence Enhancer	Adds speech clarity around the articulation range.	Amount: boost. Frequency: center of clarity. Width: how broad the boost is.
Warmth	Adds gentle saturation for thicker tone and harmonic color.	Amount: more warmth and density. Too much can distort.
Low-pass Filter	Cuts very high hiss or fizz.	LPF Hz: lower for more hiss control; higher for more open air.
Notch Filter	Cuts one narrow problem frequency, such as ring or feedback.	Frequency: target. Depth: cut amount. Q: narrowness; higher Q is narrower.
Parametric EQ	Boosts or cuts one adjustable frequency band.	Frequency: target. Gain: boost/cut. Q: width; higher Q is narrower.
Low Shelf / High Shelf	Raises or lowers everything below or above a chosen point.	Frequency: shelf edge. Gain: boost/cut. Use low shelf for body and high shelf for air/hiss.
Limiter	Final safety catch that prevents output from going over the ceiling.	Ceiling: max level. Lookahead: catches peaks early. Release: recovery. Soft clip: smoother peak handling.

Problem Solver

Problem	Try This First
Voice is boomy	Cut Mud, raise high-pass a little, lower Warmth.
Voice is thin	Lower high-pass, add a little Warmth, avoid cutting 125-250 Hz too much.
Words are hard to understand	Add a small Presence boost or use Podcast Clean.
S sounds hurt	Use de-esser, cut Sibilance, avoid boosting Air.
Fan noise between words	Use Expander before a hard Noise Gate. Raise threshold slowly.
Room sounds echoey	Add Echo Reducer gently; also move the mic closer if possible.
Output distorts	Lower Input Trim first, then lower Makeup Gain or limiter drive.

How To Know You Went Too Far

- If the voice pumps, lisps, or disappears between words, the gate, expander, compressor, de-esser, or suppression is probably too strong.
- If the voice sounds smaller after EQ, undo boosts first, then use smaller cuts.
- If the sound is clear alone but harsh with music, reduce Presence, Sibilance, or Air.
- If recordings clip, lower Input Trim. Fix clipping at the source before fixing tone.

Final Worship-Mix Advice

For church work, intelligibility beats drama. Make the words clear, keep the level steady, protect against clipping, and leave enough natural tone that the voice still feels human.